ARTIFICIAL INTELLIGENCE AND DNA: TRANSFORMING GENOMICS THROUGH MACHINE LEARNING

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Abstract

A DNA sequence carries hereditary information, but reading its letters does not by itself reveal what each region does. Some stretches encode proteins; others help control when genes are active, while the function of many sequences remains uncertain. A change in one base may affect a biological process, whereas another may have no known consequence. Researchers use artificial intelligence to examine these questions when the effect of a DNA sequence is unclear. The examples include DeepSEA and SpliceAI, as well as more recent systems such as Nucleotide Transformer and AlphaGenome. Two research resources are discussed too: the 1000 Genomes Project, which records genetic variation, and GTEx, which measures gene activity across tissues. In the studies reviewed here, machine-learning models identify patterns in DNA and estimate how a sequence may correspond to a biological measurement. Those estimates can help direct further work. They do not explain every genetic difference or establish a diagnosis on their own. Experimental findings and biological context remain necessary, particularly when the training and test data are considered. Because each system was developed for a particular task, its reported performance is interpreted in light of that task.

Why are researchers bringing AI into genetics?

A DNA molecule can be written as a long string of four letters: A, T, C and G. Recording the sequence is straightforward. Working out what it does takes more care. A region may encode part of a protein, influence the timing of gene activity, or have a function that has not yet been established. The sequence is the starting point, not the full explanation.

The effect of a DNA variant is often unclear when it is first identified. A change may leave the cell unaffected, alter the way a gene is expressed, or interfere with the production and processing of RNA. Its effect can also depend on the biological setting in which it occurs. Researchers therefore need evidence before deciding which variants warrant closer examination. Position gives a clue, but does not settle the interpretation.

Researchers can use machine learning to sort through these possibilities. They train a model using examples that pair DNA sequences with measured outcomes, including gene activity or RNA splicing. The model learns relationships from those examples and applies them to a sequence it has not seen before. Its estimate can help researchers choose a question to investigate or an experiment to run.

Question guiding the paper

How can artificial intelligence help scientists understand DNA, identify genetic changes and predict how those changes may affect living organisms?

How the research was carried out

The paper draws on published studies and research datasets. No new experiment was carried out. Nine papers and resources were selected because they describe AI methods for analysing DNA or contain biological measurements used in genetic research. Each system is discussed in relation to the problem it was built to solve, and its reported findings are considered in that context. Results from methods built for different tasks are not treated as direct comparisons.

The selected papers describe some current uses of AI in genetics, rather than every available system. No model was trained for this paper, and no patient records were examined. Medical claims are outside its scope. The paper reports what the cited research found and where biological confirmation is still needed.

Sources: [1-9]

A short guide to DNA and genes

Deoxyribonucleic acid, usually called DNA, is found in the cells of living organisms and stores hereditary information. Its familiar shape is a double helix, like a ladder twisted along its length. Sugar and phosphate molecules form the outer strands; paired chemical bases make up the rungs between them.

Four chemical bases make up DNA: adenine (A), thymine (T), cytosine (C) and guanine (G). In the usual pairing arrangement, A joins with T, while C joins with G. The sequence refers to the order of these bases. ATGCCATG is a short example; human chromosomes contain millions of bases.

Base	Symbol	Pairs with
Adenine	A	Thymine (T)
Thymine	T	Adenine (A)
Cytosine	C	Guanine (G)
Guanine	G	Cytosine (C)

A gene is a DNA segment that contributes to a functional product, most often a protein. DNA is packaged into chromosomes, and most human body cells contain 23 pairs. Gene activity differs among cell types and shifts with the cell's activity, which means a gene may be active in one tissue and quiet in another.

Molecule	Description	What follows
DNA	Stores hereditary information.	Part of this information can be copied into RNA.
RNA	Carries or processes a working copy of genetic instructions.	The RNA message can guide protein production.
Protein	Amino acids form a chain that folds into a functional molecule.	Proteins perform many jobs within cells.

Some DNA regions contain the instructions used to make proteins. Other regions regulate gene activity and are described as non-coding. That term describes their relationship to protein production; it does not imply that they are inactive or useless. A regulatory sequence, for instance, may help determine when a nearby gene is switched on.

A mutation is a change in the base sequence of DNA. It may replace a base, add one, or remove one; its effect depends partly on where the change occurs and what that region does. Many variants are harmless, or their effects have not been established. Others contribute to disease. For that reason, researchers interpret a variant in its biological setting rather than assuming that every difference is harmful.

Sources: [1, 2]

How can a computer learn from DNA?

Artificial intelligence (AI) describes computer-based methods that carry out tasks such as recognising patterns or making predictions. Machine learning is one area within AI. Instead of writing a separate instruction for every possible DNA sequence, researchers train a model on examples and let it learn relationships in the data.

In deep learning, artificial neural networks are built from connected layers. Information is processed at each layer before a representation is passed onward. A DNA model might, for example, learn that a short sequence pattern tends to appear near a measured biological signal. It may also detect combinations of patterns that are difficult to recognise by looking at individual bases in isolation.

Before a computer can process DNA, the letters must be represented in a suitable form, often as numbers or groups of letters known as tokens. Several newer models use transformers. In this network design, an attention mechanism weighs information from different positions in the sequence. That is useful when a regulatory region affects a gene at some distance, since the relevant bases may be far from the gene itself.

Step	Computer operation
1. DNA sequence	Receives a sequence made from A, T, C and G.
2. Preparation	Changes the sequence into numbers or letter groups the model can process.
3. Pattern learning	Links sequence patterns to biological features measured in experiments.
4. Prediction	Estimates an outcome for researchers to compare with other evidence.

A model is first trained on examples with known biological measurements. Its performance is then checked using a separate set of examples that was withheld during training. The distinction matters: strong results on sequences already seen by the model may reflect familiarity rather than an ability to generalise. Testing on held-out DNA gives a clearer indication of whether the learned relationships extend to new sequences.

The expression 'AI reads DNA' can be misleading. A model does not interpret a genome in the way a geneticist does. It calculates patterns from the examples supplied during training, and its estimates are shaped by the data, the biological task and the way performance is tested.

Sources: [3, 6-8]

Three models built for different questions

DeepSEA: looking beyond protein-coding genes

DeepSEA was introduced by Jian Zhou and Olga Troyanskaya in 2015. Their question was whether variants in non-coding DNA could change gene regulation. To train the model, they used large experimental datasets on chromatin, including measurements of protein binding and DNA accessibility. From a DNA sequence, DeepSEA predicts related features and can point to variants that may alter regulatory activity.

A variant can affect gene regulation even when the protein-coding sequence remains unchanged. DeepSEA used deep learning to analyse regulatory regions of the genome, where sequence alone often makes their biological effects difficult to determine.

SpliceAI: checking how RNA is joined

Jaganathan and colleagues introduced SpliceAI in 2019 to study RNA splicing. After a gene is copied into an initial RNA message, some segments are removed and the remaining sections are joined. The resulting RNA is then used by the cell. A DNA variant can disrupt the splice pattern and change the message that is produced.

SpliceAI predicts likely splice sites and flags DNA variants that may interfere with RNA processing. The researchers checked selected predictions against RNA-sequencing results. Of the patients whose variants were predicted to alter splicing, 21 of 28 had supporting evidence. That result supports the method in the group examined, although it does not show that every prediction will hold in other patients or biological settings.

Enformer: connecting distant parts of DNA

Published in 2021, Enformer predicts gene activity from DNA sequence. Its input covers a long stretch of DNA, which matters because regulatory regions can influence genes from a distance. Looking only at bases close to a gene could miss some of those signals. Enformer can incorporate information from regions up to about 100,000 bases away.

DeepSEA predicts regulatory features, SpliceAI estimates changes to RNA splicing, and Enformer models gene activity using longer sequence windows. Their scores cannot be treated as direct comparisons because each model was built for a different task.

Sources: [1-3]

Learning from much larger DNA collections

A newer line of research trains models on very large collections of DNA before adapting them to a particular task. During the first stage, the systems learn recurring features of sequence data. The approach resembles pre-training in language models, although DNA sequences represent biological information rather than words arranged by grammar.

DNABERT-2

DNABERT-2 was presented at ICLR 2024. Its developers examined how a DNA sequence should be broken into units for processing. Earlier models often relied on groups of a fixed length; DNABERT-2 instead forms groups based on patterns in the sequence. The paper also reported a benchmark with 36 datasets covering nine tasks, giving researchers a more consistent way to evaluate these models.

The way a sequence is divided affects the time and resources needed for training, as well as the model's results. Comparisons can also be difficult when systems process the same DNA using different units of sequence data.

Nucleotide Transformer

The Nucleotide Transformer paper describes several models trained on DNA sequences from humans and other species. The training material included 3,202 human genomes and 850 genomes from other species. The researchers trained the models to learn broad sequence features that could later be adapted to specific questions in genetics.

A large training collection does not guarantee strong performance on every specialised test, since the model still needs to be evaluated using data for that particular task.

AlphaGenome

AlphaGenome was published in Nature in January 2026. It takes long DNA sequences as input and predicts several kinds of biological information, including gene activity, RNA splicing, chromatin accessibility and binding by proteins involved in gene regulation. The system can process sequences containing up to one million DNA letters.

A long input sequence gives AlphaGenome access to information across a broad region of DNA. The model estimates biological measurements; it cannot determine what will happen in an individual patient or replace an experiment. Researchers can compare its predictions with other evidence before drawing any biological interpretation.

Sources: [6-8]

Where does the information come from?

Machine-learning models need examples to learn from, and DNA sequence data alone seldom provide every measurement needed to answer a biological question. Sequencing identifies which bases are present. Separate experiments can measure, for example, how active a gene is in a particular tissue. By bringing these data together, researchers can examine how genetic variation relates to cellular activity.

The 1000 Genomes Project

The 1000 Genomes Project brought together a reference collection of genetic variants found in people from different populations. Its 2015 report included 2,504 individuals from 26 populations and more than 88 million variants. The collection helps describe how common genetic differences are distributed. It does not contain every variant that might be found in any one person.

The GTEx project

GTEx examines gene activity in human tissues. The published GTEx v8 analysis included 15,201 RNA-sequencing samples from 49 tissues and 838 donors. Since a gene may be active in one tissue but show little activity in another, the effect associated with a DNA variant can depend on where that gene is expressed.

The two resources measure different parts of human genetics. The 1000 Genomes Project documents variation among individuals; GTEx links gene expression to particular tissues. In the GTEx v8 report, most donors were European American, which limits how confidently the measurements can be extended to other populations. The donor imbalance limits how far the GTEx results can be generalised, but it does not make the dataset unusable. The two resources should not be assumed to have been used to train every AI model discussed here, and donor composition affects how widely some measurements can be used.

Example	DNA sequence	Meaning
Reference	ATGCCATG	The starting sequence shown for comparison.
Changed	ATGTCATG	At position four, C has been replaced by T.

A researcher examining a variant near an RNA-splicing region might ask whether the change affects assembly of the RNA message. SpliceAI can estimate the likelihood of such an effect. Researchers would then need measurements from the relevant tissue or cells to check whether the predicted splice change occurs in that setting. Until those data support the prediction, the proposed effect remains unconfirmed.

Sources: [2, 4, 5]

What can we reasonably conclude from the studies?

The three specialist models were built for different tasks. SpliceAI assesses splicing, DeepSEA predicts regulatory features in non-coding DNA, and Enformer estimates gene activity across longer sequences. Nucleotide Transformer and AlphaGenome cover a broader set of questions, but their performance still needs to be judged task by task because a result on one task does not establish how well a model will perform on another.

Model	Question	What the prediction cannot establish
DeepSEA	Could a change in non-coding DNA affect regulatory features?	Whether the variant causes a disease.
SpliceAI	Could a DNA variant alter RNA splicing?	Whether the predicted effect occurs in every tissue or person.
Enformer	What gene-activity pattern may be associated with this sequence?	Whether one predicted activity level describes the full biological response.
DNABERT-2 / Nucleotide Transformer	Which sequence patterns can be learned from large DNA collections?	Whether a general-purpose model will beat specialised methods on every task.
AlphaGenome	Which molecular signals can be predicted from a long DNA sequence?	Whether the prediction is a clinical diagnosis or an experimental result.

A comparison published in 2025 tested five DNA foundation models on 57 datasets. The results differed across tasks. General-purpose models performed competitively on some variant-identification tests, while specialised systems did better on certain gene-expression and gene-regulation tasks. The biological question and the kind of evidence required should guide that choice.

The comparison found no basis for choosing a model solely by its size or release date. Performance may be high for one task and more modest for another. A fair assessment needs appropriate test data and a consistent task for comparison. Researchers should also test sequences that were kept out of training, because results on familiar examples can give an inflated impression of performance on new DNA.

What the evidence adds up to

Across the selected papers, the models identify variants for closer review and estimate relationships between DNA sequence, gene activity and RNA processing, narrowing the explanations researchers need to test. The human genome has not been fully interpreted by these methods. A prediction remains one piece of evidence, which needs to be considered alongside the biological setting and follow-up measurements. Its practical use is to direct experiments toward a possibility that can be tested.

Sources: [1-3, 6-9]

Useful tools, but not final answers

Where AI is useful

A computer can examine sequence information at a scale that would be difficult to inspect base by base, and models can detect patterns associated with measured biological features in experiments. The systems discussed here predict gene regulation, RNA splicing and gene activity. A variant-scoring model may help prioritise candidates; an RNA-splicing model can identify changes to check against RNA-sequencing data. Laboratory tests can then check whether the predicted effect is present, which the model alone cannot establish.

What still makes the work difficult

Predictions can be wrong, particularly when the training data do not represent the population, tissue or biological condition under study. Training examples are uneven because some features are easier to measure than others. A strong result still needs independent testing, followed by evidence that the predicted change can be observed in cells. A computer-generated score does not prove that a variant causes disease. Many genetic conditions involve more than one base change; a person's biology may reflect the combined effects of genes, environmental influences and other factors. Held-out data give one check on generalisation, though they cannot replace independent biological validation. A predicted variant effect therefore cannot, by itself, determine that person's future health.

Privacy and fairness

Genetic information needs careful protection. A person's records can reveal details about relatives as well as about that individual, so participants should understand how their data will be used and research organisations need safeguards against unauthorised access. Research datasets differ in how well they represent populations. When a dataset contains far more examples from one population than from another, predictions may be more reliable for the population most represented in the training data. Some populations may still have too few examples in a research dataset.

What may come next

Future systems may combine DNA sequences with measurements of RNA, proteins and individual cells. Such data could help explain how gene activity changes under different conditions. Researchers may also be able to identify more clearly which sequence regions contributed to a prediction. These are possible directions, not established outcomes; each will need evaluation using suitable data and experiments.

Conclusion

The papers reviewed here describe AI as a tool for DNA research. DeepSEA, SpliceAI and Enformer show how models can be built for specific biological questions. DNABERT-2 and Nucleotide Transformer learn broader patterns from large sequence collections, while AlphaGenome predicts several molecular features within one system. Model outputs still depend on the data used for training and can be wrong. A prediction is not a medical finding. AI can help researchers identify explanations that merit testing; experiments and expert judgement are needed to determine whether those explanations fit the biology.

Sources: [1-9]

References

The short source names below match the numbers cited at the ends of each section here.

[1] DeepSEA

[2] SpliceAI

[3] Enformer

[4] 1000 Genomes

[5] GTEx

[6] DNABERT-2

[7] Nucleotide Transformer

[8] AlphaGenome

[9] DNA Benchmark

Declaration of AI Assistance

Artificial intelligence (AI) tools were used to support aspects of this research, including reviewing and refining written material, improving clarity, and organizing information during the preparation of the study.